

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 4: Quadrat Sampling: Is the Distribution of Grass and Weeds Uniform Across the School Field?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	A	B	D	B	B	A

USE AFTER: OBSERVE PHASE — Jigsaw Step 1 (expert groups): placing the quadrat at 5 random locations in their zone

Q1. A team is sampling the grass and weeds in a zone of the school field using a 1 m x 1 m quadrat. Why must they place the quadrat at random locations instead of where the grass looks nicest?

- A. So that every quadrat contains exactly the same number of plants
- B. So that the sample is not biased and represents the whole zone**
- C. So that the counting is finished faster
- D. So that only the most common weed is counted

Correct answer: B. Random placement removes bias, so the small sampled area fairly represents the whole zone. Choosing spots by eye would over- or under-estimate the population.

USE AFTER: OBSERVE PHASE — Jigsaw Step 1 (expert groups): calculating total counted, mean per quadrat and density

Q2. A team places a 1 m x 1 m quadrat at 5 random spots in a sunny zone of the school field. The numbers of weed plants counted are 12, 8, 10, 15 and 5. What is the density of weeds in plants per m²?

- A. 10 plants per m²**
- B. 5 plants per m²
- C. 15 plants per m²
- D. 50 plants per m²

Correct answer: A. Total counted = 12 + 8 + 10 + 15 + 5 = 50. Density = 50 plants / (5 quadrats x 1 m²) = 10 plants per m².

USE AFTER: OBSERVE PHASE — Jigsaw Step 1 (expert groups): estimated population = density x total area of the zone

Q3. In a damp zone of the school field, the mean number of grass tufts is 6 per 1 m x 1 m quadrat. The zone has a total area of 200 m². What is the estimated total number of grass tufts in the zone?

- A. 33 tufts
- B. 1200 tufts**
- C. 120 tufts
- D. 206 tufts

Correct answer: B. Density = 6 tufts per m². Estimated population = density x total area = 6 x 200 = 1200 tufts.

USE AFTER: EXPLAIN PHASE — Jigsaw Step 2 (home groups): question 3 on how the quadrat method gives an estimate and why it is only an estimate

Q4. A class uses quadrats to estimate that there are 1200 weed plants in a zone. Why is this number only an estimate?

- A. Weeds cannot be counted accurately because they are too small
- B. Quadrats can only be used on grass and not on weeds
- C. The area of the quadrat was not known
- D. Only a small part of the zone was sampled and the result is assumed to represent the whole zone**

Correct answer: D. Quadrat sampling counts a small part of the habitat and scales it up. Chance and the number of quadrats used mean the true total may differ.

USE AFTER: EXPLAIN PHASE — Jigsaw Step 2 (home groups): claim-evidence-reasoning statement on chart paper

Q5. Home groups compare the mean plant density of two zones on the school field: 42 plants per m² in a damp zone near a tap and 6 plants per m² in a heavily trampled zone. Which claim is best supported by this evidence?

- A. Trampling has been proved to be the only cause of the low density
- B. The distribution of plants is not uniform, because density was much higher in the damp zone than in the trampled zone**
- C. The damp zone is healthier because more plants always means a healthier ecosystem
- D. The distribution of plants is uniform, because plants grow everywhere on the field

Correct answer: B. The numbers show a clear difference in density between zones, so plants are not evenly spread. The data show a pattern but do not prove a single cause, and more plants does not always mean a healthier ecosystem.

USE AFTER: EXPLAIN PHASE — Jigsaw Step 2 (home groups): question 4 linking weeds and algae in the river to the field patterns

Q6. Along a Kenyan river, aquatic weeds and algae grow thickly in one stretch where fertiliser and sewage from nearby farms enter the water. Fish are fewer there. Which explanation fits best?

- A. Fish are fewer only because they are old and have no food
- B. Extra nutrients let algae and weeds boom, which can reduce oxygen in the water and harm the fish**
- C. More weeds means more food and oxygen, so the fish there are healthier
- D. The weeds grow thickly only because the water is the same everywhere along the river

Correct answer: B. Nutrient enrichment causes algae and weeds to boom. This can lower oxygen levels and harm fish, so more plants is not always a sign of a healthy ecosystem.

USE AFTER: END OF LESSON — Exit ticket: prompt 2 on two reasons why plant density may differ between two parts of a habitat

Q7. On the school field, plant density is much lower on a heavily used footpath than beside a tap. Which is the best explanation?

- A. Trampling and compacted soil damage plants and limit root growth, while the soil beside the tap has more water**
- B. Plants need no water, so the tap makes no difference to density
- C. Seeds cannot grow anywhere that the soil is wet
- D. Plants are spread evenly, so the difference must be due only to counting mistakes

Correct answer: A. Environmental conditions such as trampling, soil compaction and water availability affect plant populations. Plants are not spread evenly, and these factors can explain the difference in density.